

Excellent!

 $\frac{9}{9} \Rightarrow 100\%$

0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7	7.5	8	8.5	9	10
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Name:

These questions assess your understanding of the material from Chapters 22 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. A proton travels with speed $1.4 \times 10^6 \frac{m}{s}$ at an angle of 21° to a uniform magnetic field. If its path has a circular component of radius 0.23 m, calculate the magnitude of the magnetic field.

$$v = 1.4 \times 10^6 \frac{m}{s} @ 21^\circ$$

$$r = 0.23 m$$

$$\textcircled{1} v = v \sin \theta$$

$$v = (1.4 \times 10^6) \sin(21)$$

$$v = 5.02 \times 10^5 \frac{m}{s}$$

$$\textcircled{2}$$

$$r = \frac{mv}{qB}$$

$$0.23 = \frac{(1.67 \times 10^{-27})(5.02 \times 10^5)}{(1.60 \times 10^{-19}) B}$$

$$B = 2.28 \times 10^{-2} T$$

$$\text{ANSWER: } 2.28 \times 10^{-2} T$$

2. What is the magnitude of the maximum force exerted on an aluminum rod with a $0.32 \mu C$ charge that you pass between the poles of a 3.3-T permanent magnet at a speed of $4.6 \frac{m}{s}$?

$$0.32 \mu C \rightarrow 0.32 \times 10^{-6} C$$

$$3.3 T$$

$$4.6 m/s$$

$$\textcircled{1} F = q v b \sin \theta$$

$$\sin = 1$$

$$\text{at max}$$

$$F = (0.32 \times 10^{-6})(4.6)(3.3)$$

$$F = 4.86 \times 10^{-6} N$$

$$\text{ANSWER: } 4.86 \times 10^{-6} N$$

3. Find the current that must run through a current-carrying circular loop so that it produces a $54 \mu T$ magnetic field at its center, given that the radius of the circular loop is 3.2 cm.

$$54 \mu T \rightarrow 54 \times 10^{-6} T$$

$$3.2 cm \rightarrow 3.2 \times 10^{-2} m$$

$$\textcircled{1} B = \frac{\mu_0 I}{2R}$$

$$54 \times 10^{-6} = \frac{(4\pi \times 10^{-7}) I}{2(3.2 \times 10^{-2})}$$

$$I = \frac{(54 \times 10^{-6})(2)(3.2 \times 10^{-2})}{(4\pi \times 10^{-7})}$$

$$I = 2.75 A$$

$$\text{ANSWER: } 2.75 A$$

+3

4. An electron moves at $4.4 \times 10^6 \frac{m}{s}$ perpendicular to a magnetic field. The field causes the electron to travel in a circular path of radius 69. cm. What is the strength of the magnetic field?

$$4.4 \times 10^6 \frac{m}{s}$$

$$69 \text{ cm} \rightarrow 69 \times 10^{-2} \text{ m}$$

$$\textcircled{1} r = \frac{mv}{qB}$$

$$69 \times 10^{-2} = \frac{(9.11 \times 10^{-31})(4.4 \times 10^6)}{(1.60 \times 10^{-19}) B}$$

$$\boxed{B = 3.63 \times 10^{-5} \text{ T}}$$

ANSWER: $3.63 \times 10^{-5} \text{ T}$ ✓

5. The hot and neutral wires supplying DC power to a light-rail commuter train carry 920 A and are separated by 53. cm. What is the magnitude of the force between 240 m of these wires?

$$920 \text{ A}$$

$$53 \text{ cm} \rightarrow 53 \times 10^{-2} \text{ m}$$

$$240 \text{ m}$$

$$\textcircled{1} \frac{F}{l} = \frac{\mu_0 (I_1)(I_2)}{2\pi r}$$

$$\frac{F}{240} = \frac{(4\pi \times 10^{-7})(920)(920)}{2\pi (53 \times 10^{-2})}$$

$$\frac{F}{240} = 0.319$$

$$\boxed{F = 76.7 \text{ N}}$$

ANSWER: 76.7 N ✓

6. Suppose that you rub a glass rod with silk, placing a 46.-nC positive charge on it. Calculate the initial force on the rod due to the Earth's magnetic field, if you throw the rod with a horizontal velocity of $5.1 \frac{m}{s}$ due west in a place where Earth's field is due north. Recall that the magnitude of Earth's magnetic field is $50. \mu\text{T}$.

$$q = 46 \text{ nC} \rightarrow 46 \times 10^{-9} \text{ C}$$

$$5.1 \text{ m/s}$$

$$50 \mu\text{T} \rightarrow 50 \times 10^{-6} \text{ T}$$

$$\textcircled{1} F = qvB \sin \theta$$

$$F = (46 \times 10^{-9})(5.1)(50 \times 10^{-6}) \sin 90$$

$$\boxed{F = 1.17 \times 10^{-11} \text{ N}}$$

ANSWER: $1.17 \times 10^{-11} \text{ N}$ ✓

7. Find the current that must run through a long, straight wire so that it produces a $33. \mu\text{T}$ magnetic field at a distance of 15. cm from the wire.

$$33 \mu\text{T} \rightarrow 33 \times 10^{-6} \text{ T}$$

$$r = 15 \text{ cm} \rightarrow 15 \times 10^{-2} \text{ m}$$

$$\textcircled{1} B = \frac{\mu_0 I}{2\pi r}$$

$$33 \times 10^{-6} = \frac{(4\pi \times 10^{-7}) I}{2\pi (15 \times 10^{-2})}$$

$$I = \frac{(33 \times 10^{-6})(2\pi)(15 \times 10^{-2})}{(4\pi \times 10^{-7})}$$

$$\boxed{I = 24.8 \text{ A}}$$

ANSWER: 24.8 A ✓

+4

8. Calculate the radius of curvature of the path of a proton having a velocity of $1.5 \times 10^7 \frac{m}{s}$ perpendicular to a magnetic field of strength $B = 0.32 T$.

$$1.5 \times 10^7 \frac{m}{s}$$

$$0.32 T$$

$$① r = \frac{mv}{qB}$$

$$r = \frac{(1.67 \times 10^{-27})(1.5 \times 10^7)}{(1.6 \times 10^{-19})(0.32)}$$

$$r = 0.489 m$$

$$\text{ANSWER: } 0.489 m$$



9. Find the maximum torque on a 25.-turn square loop of wire that is 4.7 cm on each side that carries 4.1 A of current in a 0.90-T field.

$$25 \text{ turns}$$

$$4.7 \text{ cm} \rightarrow 4.7 \times 10^{-2} m$$

$$4.1 A$$

$$0.90 T$$

$$① \tau = N I A B \sin \theta$$

$$\text{ANSWER: } 0.204 Nm$$

$$\tau = (25)(4.1)(2.21 \times 10^{-3})(0.9) \sin = 1 \text{ at max}$$

$$\tau = 0.204 N \cdot m$$



$$A = (4.7 \times 10^{-2})^2 = 2.21 \times 10^{-3} m^2$$

+2

1. $B = 0.0228 \text{ T} \therefore r = \frac{mv}{qB}$, $m_p = 1.67 \times 10^{-27} \text{ kg}$, $q_p = 1.60 \times 10^{-19} \text{ C}$
2. $F = 4.86 \times 10^{-6} \text{ N} \therefore F = qvb \sin \theta \therefore F_{\max} = qvb$
3. $I = 2.75 \text{ A} \therefore B = \frac{\mu_0 I}{2R}$
4. $B = 3.63 \times 10^{-5} \text{ T} \therefore r = \frac{mv}{qB}$, $m_e = 9.11 \times 10^{-31} \text{ kg}$, $q_e = 1.60 \times 10^{-19} \text{ C}$
5. $F = 76.7 \text{ N} \therefore \frac{F}{\ell} = \frac{\mu_0 I_1 I_2}{2\pi r}$
6. $F = 1.17 \times 10^{-11} \text{ N} \therefore F = qvb \sin \theta$, $\theta = 90^\circ$
7. $I = 24.7 \text{ A} \therefore B = \frac{\mu_0 I}{2\pi r}$
8. $r = 0.489 \text{ m} \therefore r = \frac{mv}{qB}$, $m_p = 1.67 \times 10^{-27} \text{ kg}$, $q_p = 1.60 \times 10^{-19} \text{ C}$
9. $\tau = 0.204 \text{ N} \cdot \text{m} \therefore \tau = NIAB \sin \theta \therefore \tau_{\max} = NIAB$